

# Autoencoder for Compression

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## Abstract

This report documents the development and evaluation of convolutional autoencoders for learned compression of satellite imagery. A large, unlabelled collection of Amazon rainforest patches extracted from Sentinel-2 imagery was used for self-supervised reconstruction training. Three model families were investigated: a baseline convolutional autoencoder, an autoencoder with an explicit latent-channel bottleneck, and a residual autoencoder with uniform latent quantisation and a rate-distortion-inspired loss. The reconstructed images were subsequently evaluated in a downstream semantic-segmentation task using a separately labelled Amazon rainforest dataset. This task-oriented experiment measures whether compression preserves information that is relevant to a computer-vision model rather than only visual similarity. The results show that high-quality JPEG and WebP preserve segmentation performance substantially better than the investigated learned models. The quantised autoencoder nevertheless provides a well-defined latent representation with theoretical fixed-length compression ratios between 6:1 and 24:1, depending on the number of latent channels.

## 1 Motivation and Experimental Objective

Remote-sensing imagery is spatially large and is commonly acquired repeatedly over extensive geographical areas. Storage and transmission therefore become important practical constraints. Conventional codecs such as JPEG and WebP reduce image size through manually designed transforms, quantisation, and entropy coding. Learned image compression instead optimises an encoder and decoder directly from data. The encoder maps an image  $x$  to a latent representation  $z$ , while the decoder reconstructs an approximation  $\hat{x}$ :

$$z = f_{\theta}(x), \quad \hat{x} = g_{\phi}(z). \quad (1)$$

The main objective of this project was to investigate whether such a learned representation can reduce the amount of image data while retaining information needed by a semantic-segmentation model. The work was implemented in two Jupyter notebooks. The first notebook, `image_compressor.ipynb`, constructs and trains the compression models. The second notebook, `image_segmentation.ipynb`, fine-tunes a segmentation network and evaluates it on original, autoencoder-reconstructed, JPEG-compressed, and WebP-compressed images.

This separation also distinguishes the two learning objectives. Autoencoder training is self-supervised because the input image itself is the target. Segmentation training is supervised because every image is paired with a pixel-level mask. The final evaluation therefore tests transfer across both datasets and tasks.

## 2 Theoretical Background

### 2.1 Autoencoders and Lossy Compression

An autoencoder consists of an analysis transform  $f_{\theta}$  and a synthesis transform  $g_{\phi}$ . Compression is achieved by restricting the spatial resolution, channel count, numerical precision, or entropy of the latent representation. If the bottleneck is sufficiently constrained, the model must retain image structures that are useful for reconstruction while discarding less important information.

The simplest reconstruction objective in this work is the mean squared error

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2. \quad (2)$$

MSE strongly penalises large pixel deviations but can produce visually smooth results. The explicit-latent model therefore combines the mean absolute error and MSE:

$$\mathcal{L}_{\text{rec}} = 0.8 \|x - \hat{x}\|_1 + 0.2 \mathcal{L}_{\text{MSE}}. \quad (3)$$

A practical codec additionally requires discrete symbols. Continuous floating-point activations cannot be directly represented as a compact bit stream. The quantised model maps a latent activation from  $[-1, 1]$  to one of 256 uniformly spaced levels. Rounding is non-differentiable, so a straight-through estimator is used: the forward pass contains the rounded value, while the backward pass approximates the derivative by the identity. The loss is

$$\mathcal{L} = 0.85 \|x - \hat{x}\|_1 + 0.15 \mathcal{L}_{\text{MSE}} + \lambda \frac{1}{|z_q|} \sum_i |z_{q,i}|, \quad (4)$$

where  $z_q$  is the quantised latent tensor and  $\lambda$  is the rate weight. The final term is a simple proxy encouraging smaller latent magnitudes. It is not a learned probability model and does not directly estimate the number of encoded bits. Modern learned codecs typically optimise an explicit rate–distortion objective using a probability model and entropy coding [1].

## 2.2 Semantic Segmentation as a Task-Oriented Quality Measure

Semantic segmentation assigns a class label to every pixel. Fully convolutional networks perform this dense prediction without requiring a separate classifier for each image region [7]. In this project, a pretrained FCN-ResNet50 model was adapted to two output classes and fine-tuned on Amazon rainforest masks.

The primary evaluation metric is mean intersection over union (mIoU). For class  $c$ ,

$$\text{IoU}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c}, \quad \text{mIoU} = \frac{1}{C} \sum_{c=1}^C \text{IoU}_c. \quad (5)$$

The Dice score provides a related overlap measure,

$$\text{Dice}_c = \frac{2\text{TP}_c}{2\text{TP}_c + \text{FP}_c + \text{FN}_c}. \quad (6)$$

Unlike a reconstruction-only metric, these measures expose whether compression removes boundaries, textures, or colour differences required by the segmentation network.

## 3 Data Preparation

### 3.1 Self-Supervised Sentinel-2 Patches

The compression dataset was assembled from Sentinel-2 Level-2A imagery of the Amazon rainforest. The notebook contains an optional data-preparation pipeline that searches Microsoft Planetary Computer for scenes with less than 10% cloud cover, stacks bands B04, B03, and B02 into an RGB representation, and forms a temporal median composite. Random  $512 \times 512$  patches are then extracted from geographically larger scenes.

The local experiment data contain 331 training patches and 200 validation patches. Each stored array has shape  $3 \times 512 \times 512$ , uses `uint16`, and represents Sentinel reflectance values. During loading, values above 255 are divided by 10,000 and clipped to  $[0, 1]$ . Since no labels are required, the larger variety of scenes can be used directly for reconstruction learning.

### 3.2 Supervised Amazon Rainforest Segmentation Dataset

The downstream task uses “Amazon Rainforest dataset for semantic segmentation V2” [8]. The public dataset is derived from Sentinel-2 Level-2A RGB bands and provides  $512 \times 512$  images with associated binary masks. Its published split contains 1,123 training images, 100 validation images, and 100 test images. The notebook loader pairs images and masks, converts masks to two classes, and resizes both to  $512 \times 512$ . For the documented full notebook run, a deterministic subset of 30 training images and 15 validation images was used. This reduced experiment was sufficient to verify the complete training and comparison

workflow but limits the statistical strength of the conclusions. The compression and segmentation datasets are intentionally separate: the autoencoder learns general reconstruction from the larger unlabelled patch corpus, while the labelled dataset measures the preservation of task-relevant information.

## 4 Implemented Compression Models

### 4.1 Baseline Convolutional Autoencoder

The baseline architecture contains a configurable sequence of convolution, batch-normalisation, ReLU, and max-pooling blocks. Every encoder block halves the spatial dimensions and increases the channel count. The decoder uses transposed convolutions to restore the image resolution, followed by a sigmoid output. Eight checkpoints cover encoder depths from one to four and base channel counts of 32 and 64. This architecture was useful for studying the effect of depth, but it does not define a low-channel, quantised bottleneck. Consequently, several internal representations contain more bytes than the original 8-bit RGB image.

### 4.2 Explicit-Latent Autoencoder

The second architecture uses a convolutional stem, three stride-two downsampling stages, a  $1 \times 1$  convolution to an explicit latent width, and a symmetric decoder. Models were produced with base widths 32 and 64 and latent widths 8, 16, and 32. A hyperbolic tangent bounds the latent values, and small uniform noise can be added during training to approximate the perturbation caused by later quantisation. However, the evaluated checkpoints store and process the latent tensor as `float32`; therefore this model is a compact representation but not yet a complete compressed-file format.

### 4.3 Quantised Compressive Autoencoder

The final architecture adds residual blocks before and after every scale change. The encoder depth is four, the base width is 32, and two residual blocks are used at each stage. The latent tensor is quantised to 256 values, corresponding to an ideal fixed-length representation of eight bits per latent symbol. Nine models combine latent widths  $L \in \{8, 16, 32\}$  with rate weights  $\lambda \in \{10^{-4}, 10^{-3}, 5 \cdot 10^{-3}\}$ . They were trained for 40 epochs using Adam with a learning rate of  $10^{-3}$ , a training batch size of 4, and a validation batch size of 8.

## 5 Compression-Rate Calculation

The input is treated as an 8-bit RGB image for codec comparison, giving

$$B_{\text{input}} = 512 \cdot 512 \cdot 3 \cdot 8 \quad \text{bits.} \quad (7)$$

For every model, the latent payload is calculated as

$$B_{\text{latent}} = H_z W_z C_z b, \quad \text{bpp} = \frac{B_{\text{latent}}}{512 \cdot 512}, \quad R = \frac{B_{\text{input}}}{B_{\text{latent}}}, \quad (8)$$

where  $H_z, W_z, C_z$  are the latent dimensions and  $b$  is the number of bits used for each latent value. A ratio above 1:1 indicates compression; a ratio below 1:1 indicates expansion.

### 5.1 Baseline-Model Rates

For a baseline model with encoder depth  $d$  and base width  $B$ , the bottleneck has

$$C_z = B 2^{d-1}, \quad H_z = W_z = \frac{512}{2^d}. \quad (9)$$

These models retain float32 activations, so  $b = 32$ . Table 1 shows that none of the eight baseline checkpoints compresses the raw 8-bit RGB payload.

Table 1: Latent payload calculation for all baseline autoencoders.

| Depth | Base width | Bottleneck shape            | Float32 bpp | Input:latent ratio |
|-------|------------|-----------------------------|-------------|--------------------|
| 1     | 32         | $32 \times 256 \times 256$  | 256         | 0.094:1            |
| 2     | 32         | $64 \times 128 \times 128$  | 128         | 0.188:1            |
| 3     | 32         | $128 \times 64 \times 64$   | 64          | 0.375:1            |
| 4     | 32         | $256 \times 32 \times 32$   | 32          | 0.750:1            |
| 1     | 64         | $64 \times 256 \times 256$  | 512         | 0.047:1            |
| 2     | 64         | $128 \times 128 \times 128$ | 256         | 0.094:1            |
| 3     | 64         | $256 \times 64 \times 64$   | 128         | 0.188:1            |
| 4     | 64         | $512 \times 32 \times 32$   | 64          | 0.375:1            |

## 5.2 Explicit-Latent and Quantised-Model Rates

The explicit-latent and quantised architectures perform three spatial downsampling operations, so the latent shape is  $L \times 64 \times 64$ . Table 2 includes every explicit-latent checkpoint. Base width changes the encoder and decoder capacity but not the latent payload. Because these checkpoints use float32 latent values, their implemented representation rates are 6:1, 3:1, and 1.5:1. The 8-bit column is a hypothetical rate that would require adding quantisation to this model family.

Table 2: Latent payload calculation for all explicit-latent autoencoders.

| Base | $L$ | Latent shape             | Float32 bpp | Float32 ratio | Hypothetical 8-bit ratio |
|------|-----|--------------------------|-------------|---------------|--------------------------|
| 32   | 8   | $8 \times 64 \times 64$  | 4           | 6:1           | 24:1                     |
| 32   | 16  | $16 \times 64 \times 64$ | 8           | 3:1           | 12:1                     |
| 32   | 32  | $32 \times 64 \times 64$ | 16          | 1.5:1         | 6:1                      |
| 64   | 8   | $8 \times 64 \times 64$  | 4           | 6:1           | 24:1                     |
| 64   | 16  | $16 \times 64 \times 64$ | 8           | 3:1           | 12:1                     |
| 64   | 32  | $32 \times 64 \times 64$ | 16          | 1.5:1         | 6:1                      |

All nine quantised checkpoints use 256 levels, or eight bits per latent symbol. Their complete rate calculation is shown in Table 3. The rate weight affects training but does not change the fixed-length payload for a given  $L$ .

Table 3: Fixed-length latent rates for all quantised autoencoders.

| $\lambda$ | $L$ | Latent shape             | bpp | Input:latent ratio |
|-----------|-----|--------------------------|-----|--------------------|
| $10^{-4}$ | 8   | $8 \times 64 \times 64$  | 1   | 24:1               |
| $10^{-4}$ | 16  | $16 \times 64 \times 64$ | 2   | 12:1               |
| $10^{-4}$ | 32  | $32 \times 64 \times 64$ | 4   | 6:1                |
| $10^{-3}$ | 8   | $8 \times 64 \times 64$  | 1   | 24:1               |
| $10^{-3}$ | 16  | $16 \times 64 \times 64$ | 2   | 12:1               |
| $10^{-3}$ | 32  | $32 \times 64 \times 64$ | 4   | 6:1                |
| 0.005     | 8   | $8 \times 64 \times 64$  | 1   | 24:1               |
| 0.005     | 16  | $16 \times 64 \times 64$ | 2   | 12:1               |
| 0.005     | 32  | $32 \times 64 \times 64$ | 4   | 6:1                |

Thus the quantised models theoretically remove 95.8%, 91.7%, and 83.3% of the raw RGB payload for 8, 16, and 32 latent channels, respectively. All reported rates exclude model parameters because the same decoder can be reused for many images. They also exclude headers and do not exploit symbol probabilities. No entropy coder or serialised latent-file container was implemented, so the tables report representation rates rather than measured on-disk compression ratios.

## 6 Segmentation Experiment

The segmentation notebook creates an FCN-ResNet50 model with pretrained torchvision weights and replaces its output for two classes. All parameters are fine-tuned using AdamW at a learning rate of  $10^{-4}$  for 20 epochs. The `VisionTrainer` saves latest and best checkpoints, while `LoopEvaluator` and a shared metric accumulator calculate loss, mIoU, mean accuracy, and Dice score.

For a fair comparison, every image variant uses the same validation samples, masks, normalisation, segmentation model, and metric implementation. The baseline receives the original image. Autoencoder variants pass the image through a loaded checkpoint before normalisation. JPEG and WebP variants are encoded and decoded in memory at quality settings 35, 65, and 90. Failures are recorded per variant so that one incompatible checkpoint does not terminate the complete experiment.

## 7 Results and Discussion

### 7.1 Complete Segmentation Results

Tables 4–7 report every completed variant from the full 15-image validation run. The delta is measured relative to the original-image mIoU of 0.8412.

Table 4: Original-image and conventional-codec results.

| Input variant    | mIoU   | Dice   | $\Delta$ mIoU |
|------------------|--------|--------|---------------|
| Original images  | 0.8412 | 0.9137 | 0.0000        |
| JPEG, quality 35 | 0.7125 | 0.8320 | -0.1286       |
| JPEG, quality 65 | 0.7690 | 0.8694 | -0.0722       |
| JPEG, quality 90 | 0.8370 | 0.9113 | -0.0042       |
| WebP, quality 35 | 0.5936 | 0.7439 | -0.2476       |
| WebP, quality 65 | 0.7249 | 0.8404 | -0.1163       |
| WebP, quality 90 | 0.8316 | 0.9080 | -0.0096       |

Table 5: Results for all baseline autoencoders.

| Depth | Base width | mIoU   | Dice   | $\Delta$ mIoU |
|-------|------------|--------|--------|---------------|
| 1     | 32         | 0.6014 | 0.7493 | -0.2398       |
| 2     | 32         | 0.5391 | 0.6978 | -0.3021       |
| 3     | 32         | 0.3795 | 0.5237 | -0.4616       |
| 4     | 32         | 0.3024 | 0.4353 | -0.5388       |
| 1     | 64         | 0.6978 | 0.8212 | -0.1434       |
| 2     | 64         | 0.4383 | 0.5964 | -0.4029       |
| 3     | 64         | 0.4623 | 0.6191 | -0.3789       |
| 4     | 64         | 0.5152 | 0.6750 | -0.3260       |

Table 6: Results for all explicit-latent autoencoders.

| Base width | $L$ | mIoU   | Dice   | $\Delta$ mIoU |
|------------|-----|--------|--------|---------------|
| 32         | 8   | 0.2767 | 0.3714 | -0.5644       |
| 32         | 16  | 0.2680 | 0.3567 | -0.5732       |
| 32         | 32  | 0.2611 | 0.3431 | -0.5801       |
| 64         | 8   | 0.2593 | 0.3455 | -0.5819       |
| 64         | 16  | 0.2612 | 0.3435 | -0.5800       |
| 64         | 32  | 0.2620 | 0.3461 | -0.5792       |

Table 7: Results for all quantised autoencoders.

| $\lambda$ | $L$ | mIoU   | Dice   | $\Delta$ mIoU |
|-----------|-----|--------|--------|---------------|
| $10^{-4}$ | 8   | 0.2587 | 0.3420 | -0.5824       |
| $10^{-4}$ | 16  | 0.4857 | 0.6475 | -0.3554       |
| $10^{-4}$ | 32  | 0.2724 | 0.3631 | -0.5688       |
| $10^{-3}$ | 8   | 0.4733 | 0.6317 | -0.3678       |
| $10^{-3}$ | 16  | 0.2674 | 0.3555 | -0.5737       |
| $10^{-3}$ | 32  | 0.2726 | 0.3638 | -0.5685       |
| 0.005     | 8   | 0.2632 | 0.3472 | -0.5779       |
| 0.005     | 16  | 0.4028 | 0.5519 | -0.4383       |
| 0.005     | 32  | 0.4641 | 0.6233 | -0.3770       |

The original-image baseline reached an mIoU of 0.8412 and a Dice score of 0.9137. JPEG at quality 90 was almost unchanged, losing only 0.0042 mIoU. WebP at quality 90 also remained close to the baseline. This indicates that the segmentation model is robust to mild conventional compression.

The strongest learned-model result was obtained by the shallow legacy autoencoder with 64 base channels. Its mIoU of 0.6978 is substantially below the baseline and below JPEG quality 35. More importantly, this legacy model does not provide an actual storage reduction when its float32 bottleneck is counted. Its accuracy is therefore not evidence of a favourable rate-distortion trade-off.

Among the genuinely quantised models,  $L = 16$  and  $\lambda = 10^{-4}$  performed best with mIoU 0.4857. According to Table 3, this model has a theoretical fixed-length ratio of 12:1. The  $L = 8, \lambda = 10^{-3}$  model achieved a similar mIoU of 0.4733 at a more aggressive 24:1 ratio. The results do not show a monotonic relationship between latent width, rate weight, and segmentation accuracy. For example, increasing the latent width did not consistently improve the downstream task. This suggests that optimisation stability and the reconstruction objective were more influential than latent capacity alone.

The explicit-latent, non-quantised models generally performed poorly. Their reconstructions often caused the segmentation network to favour one class, which is visible in the very low IoU for the other class in the recorded per-class metrics. A pixel-reconstruction objective does not explicitly protect semantic boundaries. Colour shifts or smoothing that appear acceptable visually can therefore be destructive for a model trained on the original image distribution.

## 8 Comparison with Published Compression Tests

Published compression results must be compared cautiously because the datasets, bit-depths, codecs, and quality metrics differ. Standard learned image-compression papers commonly use natural-image datasets such as Kodak and report PSNR or MS-SSIM as a function of measured bits per pixel. In contrast, the present experiment uses Sentinel-2 imagery and reports segmentation mIoU after reconstruction. The comparison is therefore methodological rather than a direct leaderboard comparison.

Ballé et al. implemented a complete nonlinear transform codec containing quantisation, a learned probability model, and entropy coding [2]. Their measured rate-distortion curves generally exceeded JPEG and JPEG 2000, particularly under perceptual MS-SSIM evaluation. Later, Cheng et al. improved the entropy model with discretised Gaussian mixtures and attention, obtaining performance comparable to the VVC reference codec on natural-image PSNR tests [3]. At a constrained rate of 0.15 bpp, a related low-rate model reported MS-SSIM values of 0.9761 and 0.9802 on the CLIC validation and test sets, respectively [4]. These systems differ fundamentally from the current prototype because they optimise an expected code length and generate an entropy-coded bitstream. The rate term used in this project only penalises the mean latent magnitude; it cannot exploit non-uniform symbol frequencies.

Löhdefink et al. provide a more task-relevant comparison [5]. They evaluated JPEG, JPEG2000, WebP, and a GAN-based autoencoder using downstream semantic-segmentation mIoU. At 0.0625 bpp, their GAN reconstruction exceeded JPEG2000 by 3.9 absolute mIoU percentage points, even though JPEG2000 had a 5.91 dB PSNR advantage. When the segmentation model was trained on GAN reconstructions, its mIoU improved by almost 20 absolute percentage points relative to applying a model trained only on original images. This result supports two important interpretations of the present experiment. First, reconstruction PSNR and semantic usefulness need not agree. Second, the domain mismatch between original and reconstructed images can be reduced by adapting the downstream model. The current segmentation network was trained on original images and then tested on autoencoder outputs, so it did not receive this adaptation benefit.

The closest Earth-observation comparison is the recent study by Mollière et al. [6]. It compared JPEG2000 with a learned codec using a discretised Gaussian-mixture likelihood on fire, cloud, and building segmentation. Learned compression clearly outperformed JPEG2000 on the large 10-channel Sentinel-2 building dataset, but JPEG2000 remained superior on the smaller single-channel thermal datasets. Their end-to-end compression and segmentation optimisation also failed to improve over separately optimised components. For one low-quality building experiment, joint optimisation changed the result from 0.00008 bpp, 25.76 dB, and F1 0.4925 to 0.4407 bpp, 49.80 dB, and F1 0.7090. A high-quality starting point remained approximately 0.61 bpp with F1 near 0.72. The authors concluded that learned compression is most advantageous when abundant training data and spectral redundancy are available.

Compared with these studies, the present quantised models use substantially higher fixed-length rates of 1–4 bpp while producing a larger loss of downstream accuracy. This does not imply that autoencoder compression is unsuitable for Sentinel-2 imagery. Rather, it shows that an explicit entropy model, a measured

Table 8: Contextual comparison with related compression evaluations. Values are not directly comparable across datasets and metrics.

| Study            | Domain and task                           | Rate                         | Main observation                                                                                                       |
|------------------|-------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------------------------------|
| This project     | Sentinel-2 RGB forest segmentation        | 1–4 bpp                      | Quantised AE mIoU 0.2587–0.4857; JPEG/WebP quality 90 remained close to the 0.8412 baseline.                           |
| Löhdefink et al. | Automotive semantic segmentation          | 0.0625 bpp                   | GAN codec exceeded JPEG2000 by 3.9 mIoU points; training segmentation on decoded images produced a much larger gain.   |
| Mollière et al.  | EO fire, cloud, and building segmentation | variable; example 0.4407 bpp | Learned coding won on the large multispectral dataset, while JPEG2000 was stronger on smaller single-channel datasets. |
| Cheng et al.     | Natural-image reconstruction              | 0.15 bpp                     | Low-rate learned codec reached 0.9761/0.9802 MS-SSIM on CLIC validation/test data.                                     |

bitstream, greater training diversity, and decoder-domain adaptation are central parts of competitive learned compression. The strong JPEG/WebP results in this test are consistent with Mollière et al.’s finding that traditional codecs remain difficult to beat when the learned system is small or insufficiently matched to its data and task.

## 9 Limitations and Future Work

The principal limitation is the size of the downstream experiment: only 30 training and 15 validation images were selected from the larger public dataset. The values in Tables 4–7 demonstrate the implemented workflow but should not be interpreted as a definitive benchmark. Repeating the experiment on the full published split and reporting means and confidence intervals across several random seeds would strengthen the analysis.

The quantised model is also not a complete learned codec. Its magnitude penalty is only a rate proxy, and the quantised symbols are neither modelled probabilistically nor entropy-coded. A future implementation should learn a latent probability model, encode symbols with arithmetic or range coding, and measure actual byte lengths. This would permit direct rate–distortion curves against JPEG and WebP in bits per pixel.

Further improvements include training with perceptual or multi-scale losses, adding an edge-preservation term, and including the segmentation loss during compressor training. The latter would create a task-aware codec that explicitly preserves information needed for forest segmentation. Finally, reconstruction quality should be reported with PSNR and SSIM alongside downstream metrics to separate visual fidelity from task fidelity.

## 10 Conclusion

The project established an end-to-end experimental pipeline for learned satellite-image compression and downstream semantic-segmentation evaluation. Three generations of autoencoder architecture were implemented, culminating in a residual model with an explicit, uniformly quantised latent bottleneck. Theoretical fixed-length compression ratios of 24:1, 12:1, and 6:1 were obtained for latent widths 8, 16, and 32. However, the segmentation experiment showed a clear quality gap between the learned models and conventional high-quality JPEG or WebP.

The main conclusion is that dimensional reduction alone is not sufficient for effective image compression. A useful learned codec requires discrete latent symbols, an accurate rate model, entropy coding, and an objective aligned with the information that must be preserved. The notebooks provide the foundation for this next step and, importantly, evaluate compression according to its effect on a real computer-vision task.

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